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clc;
clear;
close all;

Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20 * fc;

t = 0:1/fs:4/fm;

% Message signal
m = Am * sin(2*pi*fm*t);

% Carrier signal
c = Ac * sin(2*pi*fc*t);

% Modulation index
mu = Am / Ac;

% AM signal
am = Ac * (1 + mu*sin(2*pi*fm*t)) .* sin(2*pi*fc*t);

% Demodulation using envelope detection
dem = abs(hilbert(am));
dem = dem - mean(dem);

% FFT
N = length(am);
f = (-N/2:N/2-1) * fs/N;
AM_FFT = abs(fftshift(fft(am))) / N;

% Sideband frequencies
USB = fc + fm;
LSB = fc - fm;

% Power calculations
Pc = Ac^2 / 2;
Psb = (mu^2 * Pc) / 4;

% Plot
figure('Name', 'AM Modulation', 'NumberTitle', 'off');

subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,2);

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plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,3);
plot(t,am);
title('AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,4);
plot(t,dem);
title('Demodulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,5);
plot(f,AM_FFT);
xlim([fc-3*fm fc+3*fm]);
title('AM Spectrum / Sidebands');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

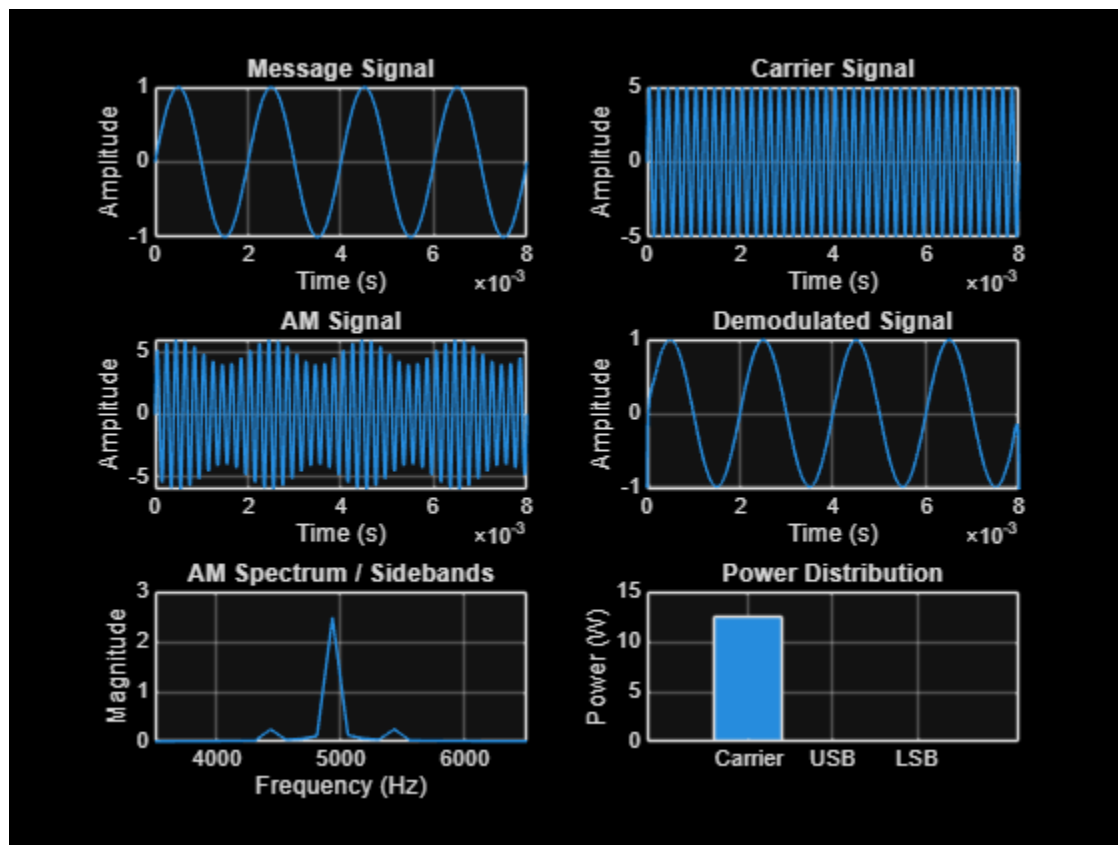
subplot(3,2,6);
bar([Pc Psb Psb]);
set(gca,'XTickLabel',{'Carrier','USB','LSB'});
title('Power Distribution');
ylabel('Power (W)');
grid on;

% Display results
fprintf('Modulation Index (mu) = %.2f\n', mu);
fprintf('USB Frequency = %d Hz\n', USB);
fprintf('LSB Frequency = %d Hz\n', LSB);
fprintf('Carrier Power = %.2f W\n', Pc);
fprintf('Each Sideband Power = %.3f W\n', Psb);
fprintf('Total Sideband Power = %.3f W\n', 2*Psb);
fprintf('Total AM Power = %.3f W\n', Pc + 2*Psb);

Modulation Index (mu) = 0.20
USB Frequency = 5500 Hz
LSB Frequency = 4500 Hz
Carrier Power = 12.50 W
Each Sideband Power = 0.125 W
Total Sideband Power = 0.250 W
Total AM Power = 12.750 W

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